Adam's original paper only used pre-training data [Penedo et al., 2024]. However, this has a problem: to predict future tokens in pre-training data, you don't necessarily need to know much about what the model is thinking, just what the prior text is.

We think that the on-policy data we use (i.e., generations from the model we are trying to interpret) are better training data because we hypothesize it to be a more solvable task, by virtue of targeting what the model is actually representing in its activations. Further, we will in practice use the AO on a model in an on-policy setting, e.g. for studying agent traces. While the above explanation is plausible, we only notice minor uplift in evaluations.

We swap FineWeb for on-policy corpus in Fig. 5; this change yields a small but measurable improvement on AObench (~$+0.03$). While the above explanation is plausible, the magnitude of the effect is modest.

3.4 Injection strength

Natural Language Autoencoders (NLAs) [Fraser-Taliente et al., 2026] inject their activations by replacing the token embedding entirely, and using a fixed scalar. We use additive, norm-matched injection after the second transformer layer following Karvonen et al. [2025].

We do not have a formal ablation for this, but on Qwen3-8B, every run that did NLA-style injection performed significantly worse than Adam's formula.

Fraser-Taliente et al. [2026] sweep their injection strength and claim that this is a quite sensitive hyperparameter. We did the same starting from Adam's formula, and found that increasing the injection strength marginally increases performance. This difference may look small, and indeed it is, but in hallucinations it is considerable (79% to 85%), which is particularly important, so we do recommend carefully choosing your hyperparameter value here.

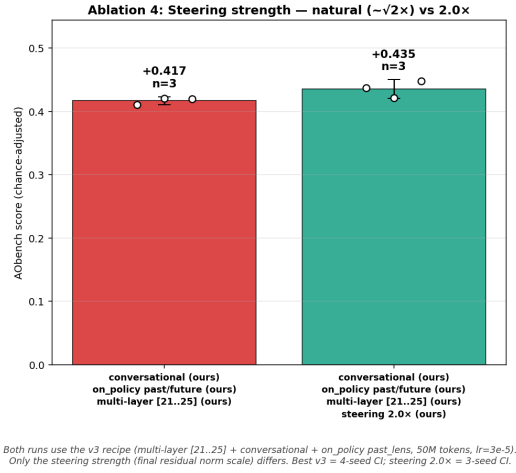


Figure 6: We ablate steering strength and find it marginally increases performance.

Our hypothesis why injecting on the second layer does better than replacing the embedding, is that the first residual stream layer has a very small cosine similarity to previous layers, a property unique to the first layer. After the first layer, cosine similarities remain fairly stable layer to layer. Because of this, it's plausible that injecting after the second layer, when the residual stream lies in the "correct basis", would work better. (Karvonen et al. [2025] reaches similar conclusions). The reason a stronger injection strength might do better is that language models have a strong prior to weight tokens sort of equally, and that it's rare that one token is load bearing for the entire explanation. Language model priors can be hard to overcome, so manually enforcing a stronger norm for the activation can help overcome this.

4 Results

We constructed AObench **with the aim of measuring what an ideal Activation Oracle should be good at**. The benchmark is a work in progress, but we recommend it as a starting point for evaluating new Activation Oracles. It targets the main frustrations identified by Jakkli et al. [2026] and reuses several model organisms from Karvonen et al. [2025]; full per-task results and prompts are reported in Appendix A.6. Concretely, *vagueness* evaluates whether the oracle's description of the model's reasoning is concrete and problem-specific, and *hallucination* evaluates whether the oracle invents specific but unsupported details about the model's reasoning.

We perform controlled ablations, starting from Karvonen et al. [2025] where we apply each of our changes. All runs are trained for exactly 50M tokens at matched learning rates, with multiple seeds where compute allowed.[4] The full recipe improves chance-adjusted AObench score from $+0.244$ (Adam baseline) to $+0.435$, with the conversational dataset swap alone accounting for the largest single jump ($+0.244 \rightarrow +0.310$).

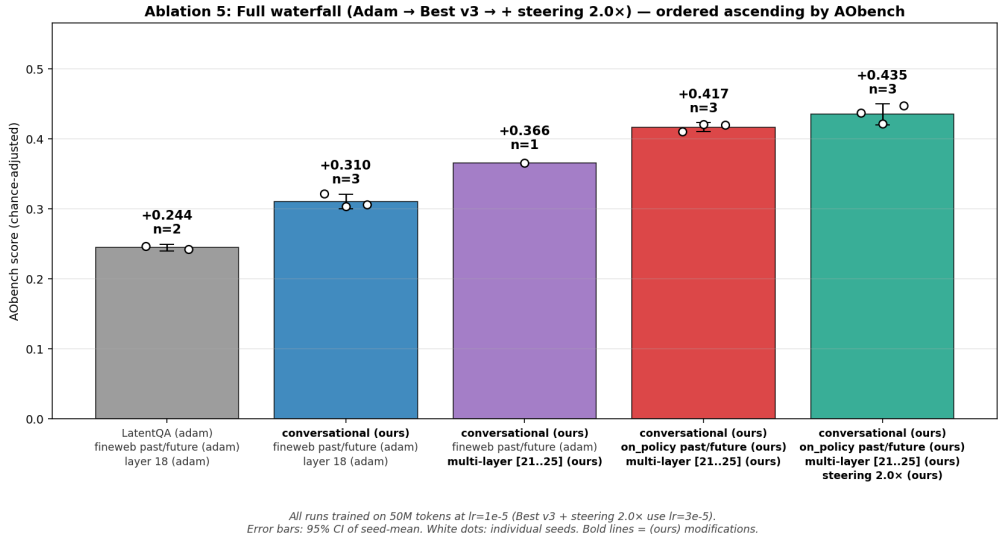


Figure 7: **AObench ablation ladder.** Each bar adds one of our interventions on top of the previous recipe. The conversational dataset swap (blue) drives the largest single-step improvement; multi-layer extraction and on-policy past/future-lens data each contribute additional uplift, and a $2\times$ injection-strength tweak yields a final small gain. All runs trained on 50M tokens; error bars show 95% CI of seed mean.

Hallucination and vagueness. Fig. 8 decomposes the ablation along the two axes most highlighted by Jakkli et al. [2026]. It's important to note that hallucination score initially increases; we attribute this to our conversational data training the model to make more specific claims, which makes it more likely for a hallucination to be counted as such. After accounting for this, the increase is monotonic (68.8% to 84.6%). Vagueness improves substantially in the full recipe relative to the original AO ($0.076 \rightarrow 0.205$ chance-adjusted), with the conversational dataset and multi-layer interventions contributing the majority of the gain.

Caveats. The on-policy past/future-lens ablation is not perfectly clean, because our conversational dataset is also constructed from on-policy chain-of-thought rollouts; some of what the on-policy lens contributes may already be expressible through the conversational data. We also caution that AOs remain difficult to evaluate: LLM judges are noisy on open-ended outputs, and several of our metrics are sensitive to prompt phrasing (Appendix A.1). With those caveats, the ordering in Fig. 7 has been stable across seeds and minor recipe variations.

[4] Our checkpoints are slightly undertrained (50M vs 65M tokens) relative to Karvonen et al. [2025]; we expect further training to improve absolute performance, but the relative ordering of recipes is already stable at our token budget (see Appendix A.2).

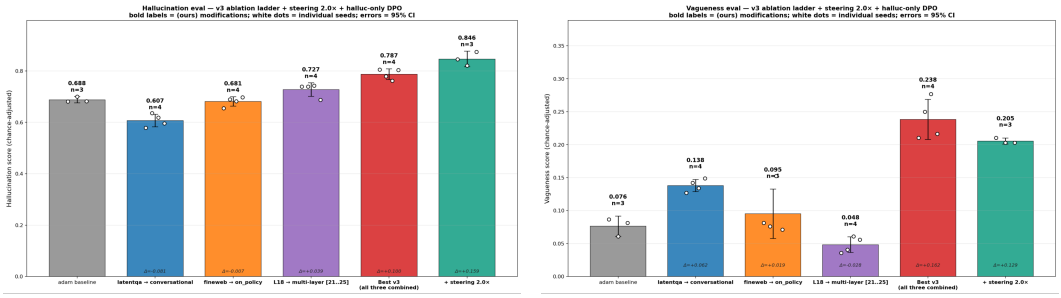


Figure 8: **Hallucination and vagueness across the ablation ladder.** Each bar adds one of our interventions on top of the previous recipe; error bars are 95% CI of seed mean.

5 Outlook

After substantial work on AOs, we believe they are a useful interpretability technique, but aren't the best tool in all circumstances. They are best used for complex open-ended questions about activations, for instance, making sense of why the model backtracked. We expect them to be particularly valuable for interpreting latent-reasoning models or any setting where substantial computation happens within a single forward pass and is therefore inaccessible to chain-of-thought monitoring. However, even with our improvements, clear limitations remain. First, they still hallucinate frequently, though this generally improves with the amount of activations supplied and uncertainty can be estimated by resampling (Appendix A.1). Second, in many settings (but not all settings) it is possible to just read the chain-of-thought directly and arrive at the same insight as the AO.

Still, we think there may be significant room for improvement by scaling up the conversational data we used, both in amount and kind. A second route is to include more narrow tasks in a "post-training" stage, though this did not yield improvements at our AO's current capability margin. Another exciting path forward is to come up with more evaluations that target something an ideal AO could plausibly solve, while being robust to text inversion concerns. If such tasks are scalable, they can be used for training as well.

More broadly, we think the idea of a scalable meta-model for activations is a promising interpretability agenda, that may scale well with model capabilities. We do not think the failures of current AOs are reason to rule out this approach.

We are particularly excited about Natural Language Autoencoders (NLAs) [Fraser-Taliente et al., 2026], and expect them to ultimately be a better way to pretrain AOs than pastlens/futurelens, since they offer a principled way to bootstrap natural-language ground truth from activations. Our observations regarding LatentQA however, remain applicable. Indeed, the original NLA paper performs its conversational finetune using LatentQA. We expect it would benefit substantially from either (i) a conversational dataset constructed along the solvability and targetedness principles introduced here, or (ii) constructing QA pairs from topics that appear in a consensus over many NLA rollouts on the same activation; essentially bootstrapping conversational solvability from bootstrapped ground truth.

We refer the reader interested in further advice for training AOs to Appendix A.2.

Contributions

Jan Bauer and Celeste De Schamphelaere contributed equally and carried out all experiments and writing. Niclas Luick came up with the idea of multilayer activation oracles, did the initial development and experimentation, and did initial exploration on quantifying uncertainty via consensus sampling. Adam Karvonen provided mentorship and guidance. Neel Nanda provided senior mentorship.

Acknowledgments

This work was conducted as part of the ML Alignment & Theory Scholars (MATS) program (cohort 10.0). We thank the MATS program for funding and compute.